

DTSA 5301

Data Science as a Field

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Core	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 2 – Vital Skills for Data Scientists	ML Engineer (70.5 / 100)	62.0 / 100
SPECIALIZATION Vital Skills for Data Scientists		

62

CURRENT AVG SCORE
across 6 job roles

89

POTENTIAL SCORE
with recommended additions

+38

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

This course provides a general introduction to the field of Data Science. It is designed for aspiring data scientists, content experts who work with data scientists, or anyone interested in learning about what Data Science is and what it’s used for. Topics include the past, present, and future of the field; the academic disciplines that both practice and make use of Data Science; collaboration between data scientists and content experts; and the practice of Data Science in the professional world. This course is part of CU Boulder’s Master’s of Science in Data Science and was collaboratively designed by both academics and industry professionals to provide learners with an insider’s perspective on this exciting, evolving, and increasingly vital discipline.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
ML Engineer	 70.5	 98.8	+28.3	MEDIUM
Applied Scientist	 63.2	 96.2	+33.0	HIGH
MLOps Engineer	 62.7	 77.7	+15.0	MEDIUM
Machine Learning Engineer	 60.9	 89.4	+28.5	MEDIUM
AI Engineer	 57.4	 95.6	+38.3	HIGH
Data Scientist	 57.3	 77.3	+20.0	MEDIUM

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

✓ Course vocabulary is well-aligned with industry job descriptions.

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS



This course is already well-aligned with industry expectations. Continue monitoring for emerging tools and frameworks.

GAP KEYWORDS BY ROLE

ML Engineer

70.5 → 98.8 (+28.3 pts)

machine models problems technical building tools

Applied Scientist

63.2 → 96.2 (+33.0 pts)

computational research interdisciplinary problems models technical regulatory feedback

MLOps Engineer

62.7 → 77.7 (+15.0 pts)

machine tools models

Machine Learning Engineer

60.9 → 89.4 (+28.5 pts)

web sets tools models applications machine technical information

AI Engineer

57.4 → 95.6 (+38.3 pts)

following models technical teams based problems machine information

Data Scientist

57.3 → 77.3 (+20.0 pts)

- sets
- problems
- approaches

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.